UNDERGRADUATE FINAL YEAR PROJECT REPORT

Department of Mathematics (BSCF)

NED University of Engineering and Technology

Determining the Efficacy of GARCH Type Models for Estimating VaR in Case of Equities Enlisted in PSX.

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Author's Declaration

We declare that we are the sole authors of this project. It is the actual copy of the project that was accepted by our advisor(s) including any necessary revisions. We also grant NED University of Engineering and Technology permission to reproduce and distribute electronic or paper copies of this project.

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Statement of Contribution

Mr. Muhammad Usman Rasheed and Ms. Maheen Badar have made a contribution to the work involving the use of R and Excel for analysis. They have accurately assessed the importance of every component and model in terms of estimating volatility and determining value at risk.

Mr. Taha Fareed and Ms. Bismah Ahmed have both contributed to the labor involved in compiling pertinent literature and creating the report.

Each of the four members has contributed and participated equally in the project's completion.

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United Nations Sustainable Development Goals

The Sustainable Development Goals (SDGs) are the blueprint to achieve a better and more sustainable future for all. They address the global challenges we face, including poverty, inequality, climate change, environmental degradation, peace and justice. There is a total of 17 SDGs as mentioned below. Check the appropriate SDGs related to the project.

• No poverty

• Zero hunger

• Good health and well being

• Quality education

• Gender equality

• Clean water and sanitation

• Affordable and clean energy

• Decent work and economic growth

• Industry, innovation and infrastructure

• Reduced inequalities

• Sustainable cities and communities

• Responsible consumption and production

• Climate action

• Life below water

• Life on land

• Peace and justice and strong institutions

• Partnerships to achieve the goal

Executive Summary

On a worldwide scale value-at-risk, or VaR, is a widely accepted methodology for evaluating risk. Financial institutions (FIs) in Pakistan employ diverse approaches to value at risk (VaR) as a result of regulatory requirements. On the contrary hand, risk takers rarely employ VaR in their decision-making processes; instead, risk controllers are the ones who measure and report VaR. In order to address this gap, this study examines several VaR strategies using a portfolio of equity assets listed on the PSX.

From January 2019 to December 2023, we acquired daily data for the KSE 30 index, and we created a hypothetical portfolio of PKR 1 billion. According to the guidelines employed by portfolio managers, the portfolio was managed by adjusting investment positions whenever a security's market value dropped below 10% of the initial investment cost. For making decisions, the much-lauded Price-to-Earnings (P/E) ratio was employed.

Various GARCH models were employed to estimate the portfolio's volatility; the most appropriate approaches were GARCH (2, 3) and GARCH (1, 2). Following that, VaR was calculated using the variance-covariance method, which considers conditional heteroscedasticity in place of static variance to account for dynamic variance. Compared to VaR calculated with static variance, our results show that VaR estimated with dynamic variance is more useful in assessing risk since it is significantly more in comparison to real historical losses and provides portfolio managers with more action points.

Acknowledgement

We extend our heartfelt gratitude to all the brilliant minds whose contributions were instrumental in conceptualizing and realizing our idea into tangible fruition.

Foremost, we express our deepest gratitude to the Almighty Allah, the creator of the universe, and the wellspring of all knowledge and wisdom. We are immensely grateful for granting us the strength, vision, expertise, and determination essential for the execution of this endeavor and for achieving the objectives of our research.

We owe a debt of gratitude to our parents whose unwavering support has propelled us to our present standing. Recognizing our moral obligation, we extend sincere appreciation to our esteemed teachers whose guidance has been invaluable throughout our journey. Special thanks are due to our dedicated supervisor, Sir Samiullah Qureshi, whose mentorship and support were indispensable in completing our research work.

We also acknowledge with gratitude the assistance rendered by other experts in Computational Finance at NED University, whose insights and expertise greatly enhanced our project.

Additionally, we express our sincere appreciation to Monis Jawed for his invaluable contributions as an Industrial Advisor.

Lastly, we wish to express our appreciation to the non-teaching staff in our department for their unwavering assistance during both the duration of this project and our tenure at the university.

Table of Contents

Author's Declaration 2
Statement of Contribution 3
United Nations Sustainable Development Goals 4
Executive Summary 5
Acknowledgement 6
Similarity Index Report 9
Abbreviations 10
CHAPTER 01 11
INTRODUCTION 11
1.1 Overview of the Equity Market 11
1.2 Overview of Risk Management in Terms of Pakistan Stock Exchange (PSX) 12
1.3 Overview of Value at Risk (VaR) and Its Methodologies 12
1.4 Overview of Volatility 14
1.5 Problem Identification 14
1.6 Scope 14
1.6 Purpose of Study 15
CHAPTER 02 16
LITERATURE REVIEW 16
CHAPTER 03 18
Conceptual Framework of Time Series 18
3.1 Analysis and the Concept 18
3.2 Time Series Models 20
3.3 Volatility Modeling: ARCH and GARCH 22
CHAPTER 04 24
ESTIMATION OF VALUE AT RISK 24
4.1 Data Collection 24
Table 1: 4.1 Data Collection 24
4.2 Initial Calculations 25
Table 2: 4.2 Initial Calculations: 26
4.2.1 Check the change 26
4.2.2 P/E Ratio: 26
4.2.3: Action Required 27
4.2.4: Index Weightages 27
4.2.5: Call to BUY or SELL 28
4.3 Portfolio Construction 28

Table 3: 4.3 Portfolio Construction 28
4.3.1 Total Cost of Investment 28
4.3.2: Available Limit 29
4.3.3: Market Value 29
Table 4.3.4: Log Returns 30
4.4 Stationarity Test/ Unit Root Test 30
Table 4: 4.4 Stationarity Test/ Unit Root Test 30
4.5 Calculation of Volatility & VaR 31
Table 5: 4.5 Calculation of Volatility & VaR 31
CHAPTER 05 35
EMPERICAL RESULTS 35
5.1 Descriptive Analysis 35
5.2 Encompass Details Regarding Volatility Estimation 37
5.3 Estimation of VaR 39
CHAPTER 06 41
CONCLUSION 41
RECOMMENDATIONS 42
REFERENCES 43
APPENDIX 44
01: Summary of GARCH (1, 0) on log return of Market Value 44
02: Summary of GARCH (2, 0) on log return of Market Value 44
03: Summary of GARCH (3, 0) on log return of Market Value 45
04: Summary of GARCH (1, 1) on log return of Market Value 45
05: Summary of GARCH (1, 2) on log return of Market Value 46
06: Summary of GARCH (2, 1) on log return of Market Value 46
07: Summary of GARCH (2, 2) on log return of Market Value 47
08: Summary of GARCH (3, 1) on log return of Market Value 47
09: Summary of GARCH (1, 3) on log return of Market Value 48
10: Summary of GARCH (1, 3) on log return of Market Value 48
11: Summary of GARCH (3, 2) on log return of Market Value 49
12: Summary of GARCH (3, 3) on log return of Market Value 49

Similarity Index Report

The following students have compiled the final year report on the topic given below in Partial fulfillment of the requirement for bachelor's degree in Computational Finance.

Project Title: **Determining the Efficacy of GARCH Type Models for Estimating VaR in Case of Equities Enlisted in PSX.**

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This is to certify that the Plagiarism test was conducted on complete report, and overall Similarity index was found to be less than 20%, with maximum 5% from single source, as Required.

Signature and Date

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Mr. Samiullah Qureshi

Abbreviations

ARCH-Auto Regressive Conditionally Heteroscedastic

VaR- Value at Risk

PSX-Pakistan Stock Exchange

IGARCH-Integrated Generalized Autoregressive Conditional heteroscedasticity

GARCHM- Generalized Auto Regressive Conditional Heteroscedasticity-in-Mean

FI GARCH-Fractionally Integrated Generalized Autoregressive Conditional Heteroscedastic

P/E: Price per Earning

EPS: Earning per Share

CHAPTER 01

INTRODUCTION

1.1 Overview of the Equity Market

Equity market, often known as the stock market, works as a key platform where, selling and purchasing of shares of publicly traded companies take place. This market plays a vital role by authorize firms to raise capital through the issuance of shares, Besides that it also allow investors to be a part of this company and earn profit based on their overall performance.

Share is considered as core of the equity market. A share denotes a possession in a company, reflects a claim of the company's assets and earnings.

Stock exchanges use as a medium in equity transactions, which are organized and regulated marketplaces where securities are traded. PSX is a prominent examples of stock exchange. Companies have to go through the process of Initial Public Offering (IPO), it is a process that permits a private firm to sell their shares publicly.

There is a vast list of participants included in the equity market, which includes retail investors—individuals who sell and purchase shares for personal use—and it also includes institutional investors, named as convention and Islamic banks, mutual funds, pension funds, and insurance companies, typically considered as the large investors in equity market There are KSE 30 and KSE 100 indices represents the performance of top 30 and top 100 companies, which provide insights into the overall market or specific sectors..

Usually shares are categorized as common shares and preferred shares. Common shares offers shareholders with voting rights and the power to receive dividends. Preferred shares, normally do not carry voting rights but provide a higher claim on assets and profits, with dividends usually being paid out before common stakeholders.

There are multiple factors that affect share prices, which includes company performance, economic indicators, market sentiment, and the basic principles of supply and demand. Company performance encompasses financial health, earnings reports, and growth prospects, while economic indicators include factors like inflation rates, interest rates, and overall economic growth. Market sentiment shows the behavior and perception of stakeholders, influenced by news, trends, and geopolitical events. The interplay of supply and demand also

plays an important role in determining share prices, with the number of shares available for sale versus the number of investors willing to buy impacting the market value.

We can say, the equity market is a dynamic and complex system that helps to raise capital for companies and provide investment opportunities for individuals and institutions.

1.2 Overview of Risk Management in Terms of Pakistan Stock Exchange (PSX)

Risk Management involves identifying, assessing, and prioritizing risks followed by coordinated efforts to minimize, monitor, and control the probability or impact of unfortunate events. Effective risk management helps investors safe their investments from uncertain events and maximize profits in a volatile market.

There are multiple risk involves in the stock market such as market risk, liquidity risk, operational risk, regulatory risk, and foreign exchange risk. Market risk involves interest rate risk, credit risk and liquidity risk. Borrowing cost can be increased due to rise in increase interest rate. If a company is enlisted in PSX, and their shares are traded publicly, fails to meet its financial obligations, will impact on stock prices and investor returns, that particular risk is named as credit risk. Liquidity risk refers to the inability to sell or purchase a stock within a particular period of time without significantly affecting its price. Regulatory risk involves the risk that changes in laws and regulations affect the business, sector or market. Foreign exchange risk refers to risk of loss occurred due to change in exchange rate between currencies.

Investors use to various techniques to mitigate these risks. Investors usually diversify their portfolio through different sectors and asset classes to minimize the effect of a below average performance of any specific asset class. Hedging techniques are also used to mitigate the effect of risk, it includes future, options and other derivative techniques. Regular monitoring of the market condition and accessing their trends helps to makes the possible changes in the portfolio.

1.3 Overview of Value at Risk (VaR) and Its Methodologies

Value at Risk (VaR) is a vital risk management tool that measures the potential loss that can be occurred in a specific period of time in portfolio. It shows that it gives exact monetary amount of loss on the portfolio that can be occurs due to market volatility under normal conditions. Calculation of VaR is really important for financial institutions, portfolio managers,

and individual investors to estimate and manage the risk linked with their portfolio, which helps investor to access situation and make proper decisions regarding their investment decisions.

VaR includes multiple important factors, including the time horizon, confidence level, and loss amount. The time horizon reflects the time period over which the risk is estimated, it usually estimated from 1 day to several months. The confidence level shows the chances that value of VaR will be accurate up to certain percent, widely used confidence levels are 95%, 90%, and 99%.

VaR is widely calculated using three different methodologies i.e. historical simulation, variance-covariance, and Monte Carlo simulation. In historical simulation method historical market data is commonly used to estimate potential losses assuming future performance will be based on past performance. First step is to collect historical data of returns, then portfolio returns are calculated for each period, sorting the returns from lowest to highest, and then at last estimating the VaR at a given confidence level.

The variance-covariance method based on the assumption that returns are normally distributed, and it uses mean and standard deviation to evaluate the VaR. First step is to collect historical data of returns, then portfolio returns are calculated for each period,, determining the potential loss at the given confidence level using the properties of the normal distribution, and calculating VaR using the formula: $VaR = Z * \sigma * \sqrt{t}$, where Z is the Z-score corresponding to the confidence level, σ is the standard deviation, and t is the time horizon.

In Monte Carlo simulation, large number of simulations of future prices are generated using computer, creates hypothetical scenarios for assets prices based on their statistical properties. The process begins with specifying key statistics such as (mean, standard deviation, correlations) of the asset returns, Multiple scenarios are used to estimate VaR value, in this way various scenarios and their VaR is predicted which helps investors to understand the dynamic market conditions and taking informed decisions.

VaR offers several benefits, including providing a clear and concise value of risk which is easy to understand and interpret. This tool is being applicable across different asset classes and portfolios. Although there are some limits of VaR. It assumes ordinary market conditions, Values will be underestimating if market dynamics goes through any stressed or unexpected situation. Another side effect is that VaR estimated through historical simulation depends on

the previous data, assuming that future values will be based on past events and the market conditions will remains unchanged. VaR estimated through Variance Co-Variance method assumes that data is normally distributed.

1.4 Overview of Volatility

Volatility is a statistical measure that assesses the extent of fluctuations in the value of a financial asset or market index over a specific period. It is typically calculated using standard deviation or variance of returns. Volatility indicates the level of uncertainty and risk associated with an asset's price movements, with higher volatility signifying greater price swings and lower volatility indicating more stable prices. Various factors contribute to market volatility, including economic indicators, geopolitical events, interest rate changes, and shifts in investor sentiment. Understanding volatility is vital for investors, as it impacts option pricing, risk management strategies, and overall portfolio performance. The Volatility Index is a widely used tool to measure and predict market volatility, providing valuable insights for investors to make informed decisions and navigate market fluctuations effectively.

1.5 Problem Identification

PSX is considered as a complex equity market as it is affected by various factors such as Geo-political events, economic policies and microstructure characteristics. These factors enhance the challenges for estimating stock market. VaR can be deployed for any portfolio in multiple ways and various approaches and there are many ways to estimate VaR. In the presence of multiple techniques, it is important to figure out the most suitable approach for measuring and reporting the risk along with creating a good buffer such that the capital of the individual financial institutions.

1.6 Scope

Pakistan is an under-developed economy and it poses a primitive financial infrastructure as compared to other countries, there exist a variety of complex financial products which provides various techniques to solve complex problems. In Pakistan, SBP works as a regulatory authority for financial institution, SBP have proposed certain methods to calculate VaR of invested portfolios which every financial institution have to follow and currently industry is using historical method to calculate VaR as supposed by the SBP. Although, VaR is an important measure for an institution and only depending on a particular method can lead to problem. We supposed to use different methods for calculating VaR and study on the nature of

calculated results using different approaches such as variance co-variance method and Monty Carlos stimulation method while using conditional variance such as auto regressive moving average models. It would lead us to predict measures such as VaR through different approaches, which will certainly be helpful in decision making for financial institutions.

1.6 Purpose of Study

GARCH model is widely used in Finance field for analyzing and forecasting the uncertainty that correlates with the risk management and investment decisions.

This study helps to understand the performance of different models of GARCH such as (E-GARCH), (T-GARCH) in estimating VaR of equities.

Historical data and back testing methodologies in GARCH lead to capture the volatility dynamics and providing reliable VaR estimation.

Also, this study will provide insights into the strengths and limitation of different GARCH-type models in terms of PSX equities that can help institutions, investors and portfolio managers about the most effective method for estimation of VaR, which can be helpful for better decision making in equity market.

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CHAPTER 02

LITERATURE REVIEW

Lim and Sek (2013) compared the performance of GARCH type models for capturing volatility in Malaysian Stock Market. They collected data for Malaysian market ranging from Jan 1990 to Dec 2010. The Malaysian market faced crisis during 1997. Therefore, data collected is divided into three parts, i.e., pre-crisis, during crisis and post crisis. The researched concluded that for asymmetric and symmetric GARCH models, both models perform differently in different time frames. During crisis, asymmetric GARCH performed well, whereas during normal period the results from symmetric GARCH were preferred. The parameters affecting volatility were exchange rates and crude oil prices, but theses parameters were insignificant during crisis.

Angelidisa, Benosa, and Degiannakis (2004) uses extended forms of ARCH Models for daily VaR calculations in a diversified portfolio comprising of five stock indices, having no. of sample sizes and distributions. The main findings concluded that firstly, leptokurtic distributions (meaning that they have many outliers), gained ground on calculated VaR forecasting. Secondly, having accurate choice of sample size is a must for VaR accuracy. And at last, they concluded that the ARCH model having forecasting of VaR with precision is different for every portfolio.

Muhammad Asif and Abdul Aziz (2016), investigates the efficiency of GARCH type models for predicting stocks return dynamic and volatility for PSX. This case study is closely related to our investigation as it focuses on the valuation of share price. Their study helps the capital analysts to identify the factor affecting the share price dynamics. They worked on a daily time series data from KSE-100 listed in PSX dated from January'1st 2008 to December'31st, 2015. They worked with ARCH family model to estimate the cluster volatility. The main finding from the study indicates that GARCH (1, 1) is the most appropriate model, indicating that the PSX market is weak-form of efficiency (meaning that the future prices are random and not influenced by the past).

Aleksandra Helena Pasieczna and Magdalena Joanna Szydłowska (2021) talks about the precision and accuracy aspect, focusing on model risk for calculating VaR at different

confidence level. This work is done on four leading European banks i.e. BNP Paribas, Credit Agricole, Commerzbank and Duetsche Bank, and the data is collected from year 2002 and 2019 from the official websites of the banks. The models that are used for this testing are Monte Carlo Technique, the historical simulation, and GARCH. The model risk is estimated by calculating the spread between the maximum and minimum VaR. The result was obtained by changing the input parameter whereas the main parameter for the estimation was time window i.e. the time interval of 125, 250, 500 days. From the study it is concluded that the accuracy alone is not enough to judge the model precision is also required.

Arum Kingsley and UchePeter (2019) discuss volatility modeling using ARCH and GARCH models in Nigerian stock market. They collected the data of almost five years from 4th of March 2011 to 31st of December 2015 excluding weekends and public holidays. This research indicates that GARCH (0,3) and GARCH (1,1), is the best model for that capture the volatility of insurance stock according to the information of Akaike, Bayesian, Shibata and Hanna Quinn. Also, it shows that VaR estimated at 95% confidence level determines the maximum expected loss in the insurance stocks on daily basis.

Anton Sorin Gabriel (2012) aims to evaluate the forecasting potential of GARCH models in the estimation of in-sample and out of sample forecasting accuracy of Romanian market from 09/03/2001 to 02/29/2012. According to their findings TGARCH is considered as the best model for forecasting the Volatility of Romanian index.

CHAPTER 03

Conceptual Framework of Time Series

3.1 Analysis and the Concept

Time series analysis is an important aspect for statistical methodology when dealing with data recorded at a consecutive time (in even intervals). Whereas, other data types of time series data that are sorted in chronological manner. Due to this ordering, we can observe an essential pattern including trends cycles, and seasonality. These are important for better understanding and predicting the future behavior for the provided data. Trend signifies a sustained change in the series, either upward trending, downward trending or constant, reflecting the overall shift in which data is moving. Cyclic patterns, transpires at irregular duration impacted by the economical and environmental factors. These irregular durations signify the arbitrary deviation, not showing any pattern or trend and are volatile. Seasonality, unlike cyclic, pertains to periodic variations transpiring at regular durations, including monthly, quarterly, caused by seasonal aspect.

The time series analysis implementation extends along multiple fields, involving economics, finance, engineering and environment science. Let’s consider, in economics, it’s utilized for predicting the GDP, unemployment rates, inflations rates, etc. For the field of finance, time series analysis helps in examining the stock prices, interest rates, exchange rates etc. In environmental science, time series is used to study about climate changes, water levels etc., whereas in engineering, it monitors and analyzes the system performance.

Stationarity is a core idea in time series analysis, plays a vital role in modelling and forecasting. It portrays a series with statistical concepts, for instance mean, variance, autocorrelation, etc. Stationarity is a fundamental concept in time series analysis, crucial for accurate modeling and forecasting. It describes a series whose statistical properties, such as mean, variance, and autocorrelation, stay unchanged overtime. There are three types of stationarity:

Strict (or strong) stationarity refers to collective distribution of any set of data points that remains consistent regardless of time shifts. It's the most severe form of stationarity.

Weak (or second-order) stationarity refers to the mean of a data not changing over time neither its variance. It's less complex than strict stationarity but vital for various time series models.

Trend stationarity implies that a series can be converted to stationary by eradicating a fixed trend component. It concedes that while the series itself may not be stationary, extracting a predictable trend can render it.

Foremost element for the selection of relevant modelling approach and assuring predictable predictions to comprehend which type of stationarity is applicable on the provided time series data, leading to testing for stationarity, which is an essential step in time series analysis.

Some common tests include the ADF test (Augmented Dickey-Fuller), which evaluates the existence of a unit root signifying non-stationarity,

$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \delta \Delta y_{t-1} + \ldots + \delta_{p-1} \Delta y_{t-p+1} + \varepsilon_t$

Where,

y_t is the time series,

Δy_t is the differenced time series,

t represents the time

α, β, γ and δ_p are coefficient to be estimated

The null hypothesis of the ADF test is typically that the time series has a unit root (indicating non-stationarity).

Second, the KPSS test (Kwiatkowski-Phillips-Schmidt-Shin), which demonstrates the null hypothesis of stationarity.

KPSS = $\frac{\sum_{t=1}^{T} S_t^2}{\sigma^2}$ where,

S_t are the cumulative sum of the deviations from the mean and,

σ^2 is the est. variance of process.

The null hypothesis of the KPSS test is that the time series is trend-stationary (weak stationarity).

Perron (PP) test can be applied also, it's similar to the ADF test but it consists less assumptions about the data generation.

There are several methodologies to attain stationarity, if the time series is non stationary. Firstly we have differencing method, this requires subtracting the previous observation from the current. It's a commonly used method.

Next, we have detrending method which removes deterministic trend components or exponential trends, and seasonal adjustment. This is also an effective method for transforming a non-stationary series into a stationary one. Achieving stationarity is important for building reliable time series models and making accurate forecasting.

3.2 Time Series Models

Several models are implemented in time series analysis to comprehend and forecast future values based on earlier observed data. Expected value models, Moving Averages (MA), Autoregressive Moving Averages (ARMA), and Autoregressive Integrated Moving Averages (ARIMA), comprise a few of those models.

3.2.1 Expected Value Models

An essential concept in statistical and probability theory is that the expected value, also additionally referred to as the mean, is the average or median value of an uncertain variable. The expected value in time series analysis assists to determine the central trend that the data moves around. Expected value models presume that, after incorporating evident trends and shifts the series' predicted future values will likely fall within a range around the historical mean.

3.2.2 Auto Regressive (AR) Models

AR models utilize a linear combination of past trends in order to predict future values. The AR (p) model, where "p" denotes the number of lagged observations, can be represented as:

$y_t = C + \emptyset_1 y_{t-1} + \emptyset_2 y_{t-2} + \ldots + \emptyset_p y_{t-p} + \varepsilon_t$ where,

y_t is the value of the time series at time.

C is a constant.

$\emptyset_1, \emptyset_2$ and $\emptyset_p$ are coefficients that determine the relationship between the current value and its p lagged values.

ε_t is the error term (white noise) at time.

3.2.3 Moving Average (MA) Models

MA models employ past errors in forecasting to estimate future values. MA models function efficiently at capturing random noise or shocks in the data. The number of lag forecast errors are represented by "q" in the MA (q) model, which may be expressed as follows;

$y_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \cdots + \theta_p\varepsilon_{t-p}$where,

$\mu$is the mean of the series,

θ are the parameters,

ϵt are the error terms

3.2.4 Auto Regressive Moving Average (ARMA) Models

For the purpose to adequately represent both momentum and shock aspects in the data set, ARMA models integrate the AR and MA models. Regarding series that demonstrate both unpredictability and serial correlation, these models are effective. One way to depict the ARMA (p, q) model is as follows,

$y_t = C + \emptyset_1 y_{t-1} + \emptyset_2 y_{t-2} + \ldots + \emptyset_p y_{t-p} + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \cdots + \theta_q\varepsilon_{t-q} + \varepsilon_t$

Where p is the order of the autoregressive part and q is the order of the moving average part.

3.2.5 Auto Regressive Integrated Moving Average (ARIMA) Models

By incorporating a differentiation phase, ARIMA models expand upon ARMA models to deal with non-stationary series. For the purpose of keeping the series stationary, "d" differencing steps are incorporated into the ARIMA (p, d, q) model. It can be represented as;

$$\nabla^d y_t = C + \emptyset_1 \nabla^d y_{t-1} + \emptyset_2 \nabla^d y_{t-2} + \cdots + \emptyset_p \nabla^d y_{t-p} - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \cdots \theta_p \varepsilon_{t-p} + \varepsilon_t$$

Where ∇^d indicates differencing "d" times. ARIMA models are versatile and widely used for forecasting purposes.

3.3 Volatility Modeling: ARCH and GARCH

During time series financial analysis, volatility—that is, a time series' frequency of variation—is crucial. Models for modelling and forecasting volatility include GARCH (Generalized Auto Regressive Conditional Heteroscedasticity) and ARCH (Auto Regressive Conditional Heteroscedasticity) models.

3.3.1 Auto Regressive Conditional Heteroscedasticity (ARCH) Models

Engle in 1982 introduced ARCH models, the variance of the current error term has been modelled as a function of the previous error terms. The model ARCH (q) can be expressed as:

$$\sigma^2_t = \alpha_0 + \alpha_1 \varepsilon^2_{t-1} + \alpha_2 \varepsilon^2_{t-2} + \cdots + \alpha \varepsilon^2_{t-q}$$

Where ε_t is the error term,

σ^2_t is the conditional variance of y_t at time t,

α are the parameters.

ARCH models are useful for capturing periods of volatility clustering in financial time series.

3.3.2 Generalized Auto Regressive Conditional Heteroscedasticity (GARCH) Models

Bollerslev in 1986 proposed GARCH models, which constitute a modification of ARCH models with lagged conditional variance values. One way to depict the GARCH (p, q) model is as follows:

$$\sigma^2_t = \omega + \sum_{i=1}^{p} \alpha_i \varepsilon^2_{t-i} + \sum_{j=1}^{q} \beta_j \varepsilon^2_{t-j}$$

Where β are the parameters for the lagged conditional variances.

Financial time series that display volatility clustered might be well predicted using GARCH models, which are extensively employed in risk management and financial forecasting.

CHAPTER 04

ESTIMATION OF VALUE AT RISK

This process involves several steps based on established theories. Here's a breakdown of what those steps are:

4.1 Data Collection

This study uses PSX website as a primary data source. It includes daily share prices of KSE 30 companies starting from 1st-January-2019 till 30th -December-2023. In second step, calculated the change in prices. KSE 30 Index weightages listed in the PSX website. EPS (Quarterly) is fetched from each company's report.

Table 1: 4.1 Data Collection

DATE	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL
1-Jan-19	134.07	22.95	23.41	42.66	231.60	249.92	123.25	71.81	58.02	8.37	154.99	204.28	146.29	83.98	80.26	128.65	72.22
2-Jan-19	134.50	22.79	23.40	43.19	230.62	244.08	123.14	72.60	58.03	8.00	154.16	196.12	138.98	82.67	77.51	129.71	75.09
3-Jan-19	129.98	22.19	22.34	42.59	228.82	241.35	123.51	72.21	58.61	8.00	149.96	192.29	132.04	82.31	76.01	131.56	75.00
4-Jan-19	130.18	22.36	22.08	42.75	227.40	240.84	125.41	72.02	57.83	8.00	150.08	189.15	132.12	82.20	75.38	131.39	75.50
7-Jan-19	135.10	22.58	23.15	44.87	235.80	244.53	129.60	74.86	58.09	8.41	156.69	184.47	133.57	85.15	79.06	137.95	78.64
8-Jan-19	139.14	23.65	24.28	46.51	238.69	248.13	130.42	76.15	58.45	8.41	164.19	182.17	139.13	86.61	83.01	137.76	79.15
9-Jan-19	139.62	24.83	24.24	45.64	236.17	245.13	131.46	75.17	58.80	8.20	165.71	178.11	138.10	85.73	84.67	136.80	78.36
10-Jan-19	142.96	24.15	23.72	44.83	235.51	242.11	132.02	74.98	58.73	7.80	171.16	179.00	131.20	84.81	85.43	138.34	78.77
11-Jan-19	142.40	24.21	23.26	44.13	235.42	254.21	132.93	75.70	57.95	8.00	171.88	177.61	131.25	83.89	84.07	137.92	78.53
14-Jan-19	142.06	24.51	24.42	45.81	238.37	264.41	134.25	75.35	58.15	8.00	174.66	183.57	132.88	85.97	85.19	140.12	78.00
15-Jan-19	146.23	24.33	24.31	44.28	238.34	256.13	134.63	74.55	58.16	7.80	183.14	184.13	130.83	84.78	85.83	141.00	78.08
16-Jan-19	144.91	24.10	23.80	43.30	236.25	248.61	135.31	74.25	57.82	7.90	180.17	184.29	128.26	82.72	84.55	139.58	77.25
17-Jan-19	144.54	24.26	24.46	43.46	236.05	243.11	135.37	74.60	58.03	8.00	178.83	184.55	133.09	82.78	84.53	140.35	76.08
18-Jan-19	145.15	24.02	24.81	43.24	235.16	241.66	134.05	74.87	57.51	8.00	178.05	185.73	130.36	82.11	84.31	139.28	76.00
21-Jan-19	147.87	23.66	24.73	41.75	236.41	245.53	137.93	74.75	57.64	7.99	182.74	186.54	123.85	79.74	83.09	142.29	76.56
22-Jan-19	148.62	24.08	25.93	42.85	287.87	246.09	141.50	75.11	57.81	7.99	183.81	182.07	128.56	81.90	85.19	144.91	78.74
23-Jan-19	149.47	24.01	26.25	42.86	239.91	247.24	144.61	74.20	58.15	7.71	182.90	182.69	126.57	82.29	86.36	146.65	79.00
24-Jan-19	149.62	24.43	25.58	44.22	236.97	242.78	148.01	73.00	58.67	7.80	185.78	182.83	130.31	84.20	85.76	148.41	79.13
25-Jan-19	148.26	13.92	26.45	44.25	233.46	240.29	149.28	73.69	58.31	7.80	184.66	183.56	127.14	83.32	84.09	153.14	79.35

DATE	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL	MARI	COLG
1-Jan-19	279.47	84.27	42.61	277.76	196.09	94.04	89.83	66.36	445.09	455.06	284.21	1,153.81	1,519.20
2-Jan-19	267.46	84.61	44.74	280.68	189.92	93.75	89.83	65.00	430.29	439.36	270.19	1,169.81	1,519.20
3-Jan-19	261.41	82.92	45.20	286.35	188.62	93.31	87.93	65.47	430.25	442.20	269.03	1,186.27	1,519.20
4-Jan-19	262.24	82.31	45.86	285.05	187.98	93.01	87.93	65.23	430.09	441.71	272.95	1,182.59	1,519.20
7-Jan-19	264.46	85.98	47.02	295.09	194.23	94.35	88.43	66.35	441.88	461.29	278.16	1,222.23	1,512.30
8-Jan-19	268.83	87.57	47.24	296.35	192.95	96.29	88.87	67.49	453.97	470.39	276.75	1,262.94	1,512.30
9-Jan-19	260.82	87.22	47.01	294.74	193.06	96.42	89.20	66.78	468.80	462.13	272.73	1,258.85	1,512.30
10-Jan-19	252.77	86.54	46.88	298.57	194.97	96.24	88.11	65.98	477.87	462.36	278.05	1,262.25	1,512.30
11-Jan-19	251.76	85.37	47.40	292.11	194.58	101.05	87.93	64.67	482.24	458.65	275.42	1,238.72	1,450.40
14-Jan-19	253.43	86.40	47.57	294.95	195.82	103.91	89.15	64.25	477.68	467.68	268.41	1,224.73	1,450.40
15-Jan-19	247.45	86.47	47.74	302.73	197.53	105.48	88.88	62.57	480.83	469.08	264.60	1,214.75	1,450.40
16-Jan-19	242.36	85.27	47.05	299.28	195.52	104.25	89.31	62.19	471.92	461.15	263.41	1,187.85	1,450.40
17-Jan-19	245.76	85.41	46.93	298.63	195.38	103.75	90.45	62.68	467.88	466.37	263.75	1,202.91	1,381.70
18-Jan-19	242.11	85.90	47.40	298.01	195.79	104.99	89.35	62.57	468.57	463.44	260.41	1,204.66	1,374.80
21-Jan-19	230.01	85.93	47.81	297.98	197.38	106.81	88.43	61.42	477.54	455.99	266.08	1,207.53	1,374.80
22-Jan-19	232.91	87.44	48.29	300.84	199.61	107.23	88.84	62.06	479.27	459.22	266.65	1,204.98	1,374.80
23-Jan-19	227.14	88.45	48.37	302.85	200.22	105.54	88.55	62.10	477.40	458.85	273.81	1,206.36	1,374.80
24-Jan-19	232.96	88.47	48.70	305.69	205.41	106.08	88.17	62.95	475.43	468.36	281.81	1,211.45	1,373.40
25-Jan-19	230.11	88.98	48.67	305.80	204.96	106.58	87.19	62.61	472.59	465.19	287.76	1,214.07	1,373.40

Data of KSE-30 Index from PSX.

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL
9/30/2018 (EPS)	6.22	(1.37)	0.03	0.58	12.82	3.95	6.15	4.50	2.00	0.07	7.20	18.11	6.79	0.95	4.09	2.90	1.85	(13.33)	1.29	1.60	6.25	3.84	2.99	5.39	2.44	48.13	7.71	25.36

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL
12/30/2018 (EPS)	6.22	(1.37)	(0.25)	2.25	10.86	3.42	8.22	12.48	7.67	1.37	7.43	(10.30)	(24.42)	3.98	4.92	12.44	7.61	(34.91)	1.33	5.72	24.28	18.02	11.35	9.59	5.82	15.26	17.92	41.11

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL
3/31/2019 (EPS)	6.64	(1.54)	0.07	1.94	15.15	2.95	2.08	3.00	2.44	(0.14)	6.27	2.40	(0.98)	5.99	3.32	3.30	1.85	(16.61)	2.43	1.76	7.32	4.29	2.91	4.28	12.74	31.56	25.65	62.65

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL
6/30/2019 (EPS)	7.69	(13.68)	0.003	2.47	27.06	12.44	2.53	2.49	3.23	(0.30)	27.18	(13.53)	(50.51)	3.67	11.16	7.80	4.01	(108.70)	6.70	3.50	2.31	4.72	4.09	6.66	9.98	59.44	32.44	82.14

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL
9/30/2019 (EPS)	6.35	0.59	0.032	(2.20)	9.02	2.58	5.89	2.49	3.06	(0.37)	6.28	(8.20)	3.47	(3.26)	2.96	11.77	6.30	(8.49)	1.24	5.20	7.40	4.74	2.80	8.76	(1.92)	14.12	2.96	8.04

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL
12/30/2019 (EPS)	6.01	(6.06)	0.0012	(2.42)	13.71	2.86	10.45	12.64	11.84	(1.14)	3.79	(13.88)	(11.74)	(1.93)	4.44	15.63	10.05	(10.29)	1.10	7.15	24.83	20.23	13.45	10.95	4.22	23.42	7.16	30.16

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL
3/30/2020 (EPS)	7.08	(11.89)	0.001	(4.38)	6.41	2.84	2.79	0.43	4.28	(0.19)	5.40	(40.49)	(8.34)	(4.22)	1.71	4.13	2.57	11.15	3.06	1.59	1.35	5.50	3.35	4.09	5.63	33.74	24.31	29.67

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL
6/30/2020 (EPS)	3.82	(17.74)	0.14	(5.30)	(13.77)	11.56	10.32	2.91	4.80	(0.26)	18.47	(73.62)	26.50	(4.93)	9.46	9.31	6.50	(50.82)	7.84	3.14	7.08	12.44	7.42	7.50	(9.74)	57.69	10.34	38.36

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL
9/30/2020 (EPS)	5.45	0.49	1.50	0.28	10.96	2.79	17.17	5.37	4.53	(0.29)	5.27	(56.69)	(5.27)	(0.80)	5.01	13.13	11.82	(16.40)	2.51	4.69	7.68	19.03	12.49	6.84	1.59	12.78	6.89	20.25

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL
12/30/2020 (EPS)	4.39	0.14	6.61	1.48	20.28	4.02	21.06	18.58	15.67	(0.54)	4.37	(45.05)	(2.13)	1.81	4.32	17.07	16.03	6.11	5.44	5.89	28.29	24.50	16.36	17.31	5.81	10.64	14.04	52.94

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL
3/30/2021 (EPS)	5.61	1.01	5.14	2.59	38.86	1.21	5.68	4.30	4.31	0.01	4.37	11.17	2.06	6.50	4.76	6.05	4.14	21.44	3.25	1.95	6.22	5.73	4.57	4.21	11.44	10.32	56.36	76.72

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL
6/30/2021 (EPS)	5.86	1.52	47.40	5.69	62.07	9.11	12.04	3.57	4.60	0.63	19.27	11.07	(20.12)	8.49	17.32	6.20	8.13	22.14	16.52	1.95	10.58	6.71	2.85	9.72	16.50	47.14	43.51	103.12

Data of Earning per share from quarterly reports of the companies, obtained from PSX.

4.2 Initial Calculations

Initial calculations is based on certain assumptions, it includes that action will be required on the change if it falls down to -10% or below. If P/E ratio is greater than the average P/E ratio of the company, action of purchase of shares to the equivalent amount will happen, otherwise selling 90% holding of that particular share take place.

IF (AND (P/E ratio <AVG P/E ratio, P/E ratio>0),"Purchase equivalent to limit", "sell 90% of holding")

The P/E ratio is calculated by dividing each share price with their respective quarterly EPS fetched from each company's report.

Table 2: 4.2 Initial Calculations:

4.2.1 Check the change

=(D7-$D$6)/$D$6

DATE	OGDC	change	PRL	change	TRG	change	MLCF	change	PSO	change
1-Jan-19	134.07	-	22.95	-	23.41	-	42.66	-	231.60	-
2-Jan-19	=(D7-$D$6)/$D$6			(0.01)	23.40	(0.00)	43.19	0.01	230.62	(0.00)
3-Jan-19	129.98	(0.03)	22.19	(0.03)	22.34	(0.05)	42.59	(0.00)	228.82	(0.01)
4-Jan-19	130.18	(0.03)	22.36	(0.03)	22.08	(0.06)	42.75	0.00	227.40	(0.02)
7-Jan-19	135.10	0.01	22.58	(0.02)	23.15	(0.01)	44.87	0.05	235.80	0.02
8-Jan-19	139.14	0.04	23.65	0.03	24.28	0.04	46.51	0.09	238.69	0.03
9-Jan-19	139.62	0.04	24.83	0.08	24.24	0.04	45.64	0.07	236.17	0.02
10-Jan-19	142.96	0.07	24.15	0.05	23.72	0.01	44.83	0.05	235.51	0.02
11-Jan-19	142.40	0.06	24.21	0.05	23.26	(0.01)	44.13	0.03	235.42	0.02
14-Jan-19	142.06	0.06	24.51	0.07	24.42	0.04	45.81	0.07	238.37	0.03
15-Jan-19	146.23	0.09	24.33	0.06	24.31	0.04	44.28	0.04	238.34	0.03
16-Jan-19	144.91	0.08	24.10	0.05	23.80	0.02	43.30	0.02	236.25	0.02
17-Jan-19	144.54	0.08	24.26	0.06	24.46	0.04	43.46	0.02	236.05	0.02
18-Jan-19	145.15	0.08	24.02	0.05	24.81	0.06	43.24	0.01	235.16	0.02
21-Jan-19	147.87	0.10	23.66	0.03	24.73	0.06	41.75	(0.02)	236.41	0.02
22-Jan-19	148.62	0.11	24.08	0.05	25.93	0.11	42.85	0.00	287.87	0.24
23-Jan-19	149.47	0.11	24.01	0.05	26.25	0.12	42.86	0.00	239.91	0.04
24-Jan-19	149.62	0.12	24.43	0.06	25.58	0.09	44.22	0.04	236.97	0.02
25-Jan-19	148.26	0.11	13.92	(0.39)	26.45	0.13	44.25	0.04	233.46	0.01

Checking the change when it reaches to -10%, this is our desired point.

4.2.2 P/E Ratio:

=BR4/EPS!$N$3

		SHEL			
S.NO	DATE	NO. OF SHARES	CHANGE	Check	P/E RATIO
1	1-Jan-19	28,392	-	Ok	=BR4/EPS!$N$3
2	2-Jan-19	28,392	-4%	Ok	10.83
3	3-Jan-19	28,392	-6%	Ok	10.62
4	4-Jan-19	28,392	-7%	Ok	10.44
5	7-Jan-19	28,392	-10%	Ok	10.19
6	8-Jan-19	53,092	-11%	Action Required	10.06
7	9-Jan-19	53,092	-2%	Ok	9.83
8	10-Jan-19	53,092	-2%	Ok	9.88
9	11-Jan-19	53,092	-3%	Ok	9.81
10	14-Jan-19	53,092	1%	Ok	10.14
11	15-Jan-19	53,092	1%	Ok	10.17
12	16-Jan-19	53,092	1%	Ok	10.18
13	17-Jan-19	53,092	1%	Ok	10.19
14	18-Jan-19	53,092	2%	Ok	10.26
15	21-Jan-19	53,092	2%	Ok	10.30
16	22-Jan-19	53,092	0%	Ok	10.05
17	23-Jan-19	53,092	0%	Ok	10.09
18	24-Jan-19	53,092	0%	Ok	10.10
19	25-Jan-19	53,092	1%	Ok	10.14
20	28-Jan-19	53,092	6%	Ok	10.64
21	29-Jan-19	53,092	11%	Ok	11.17
22	30-Jan-19	53,092	16%	Ok	11.63
23	31-Jan-19	53,092	20%	Ok	12.04

Calculating P/E ratio by the help of earnings per share and share price.

4.2.3: Action Required

=IF(F6<=-10%,"Action Required", "Ok")

	B	C	D	E	F	G	H	I
						OGDC		
	S.NO	DATE	PRICE	NO. OF SHARES	CHANGE	Check	P/E RATIO	Action
	1	1-Jan-19	134.07	390,841	-	Ok	21.55	
	2	2-Jan-19	134.50	390,841	0.32%	Ok	21.62	
	3	3-Jan-19	129.98	390,841	=IF(F6<=-10%,"Action Required", "Ok")			
	4	4-Jan-19	130.18	390,841	-2.90%	IF(logical_test, [value_if_true], [value_if_false])		
	5	7-Jan-19	135.10	390,841	0.77%	Ok	21.72	
	6	8-Jan-19	139.14	390,841	3.78%	Ok	22.37	
	7	9-Jan-19	139.62	390,841	4.14%	Ok	22.45	
	8	10-Jan-19	142.96	390,841	6.63%	Ok	22.98	
	9	11-Jan-19	142.40	390,841	6.21%	Ok	22.89	
	10	14-Jan-19	142.06	390,841	5.96%	Ok	22.84	
	11	15-Jan-19	146.23	390,841	9.07%	Ok	23.51	
	12	16-Jan-19	144.91	390,841	8.09%	Ok	23.30	
	13	17-Jan-19	144.54	390,841	7.81%	Ok	23.24	
	14	18-Jan-19	145.15	390,841	8.26%	Ok	23.34	
	15	21-Jan-19	147.87	390,841	10.29%	Ok	23.77	
	16	22-Jan-19	148.62	390,841	10.85%	Ok	23.89	
	17	23-Jan-19	149.47	390,841	11.49%	Ok	24.03	
	18	24-Jan-19	149.62	390,841	11.60%	Ok	24.05	
	19	25-Jan-19	148.26	390,841	10.58%	Ok	23.84	
	20	28-Jan-19	147.59	390,841	10.08%	Ok	23.73	
	21	29-Jan-19	150.00	390,841	11.88%	Ok	24.12	
	22	30-Jan-19	150.20	390,841	12.03%	Ok	24.15	

Checking that action is required if the change is greater than 10%.

4.2.4: Index Weightages

KSE-30	OGDC	PRL	TRG	MLCF	PSO	SEARL	HBL	EFERT	MEBL	TPLP	PPL	SHEL	ATRL	DGKC	SNGP
WEIGHTAGE	5.24%	0.42%	1.85%	1.15%	2.45%	0.74%	4.24%	5.97%	6.08%	0.17%	4.87%	0.58%	1.07%	0.97%	1.20%
ALLOCATION	52,400,000	4,200,000	18,500,000	11,500,000	24,500,000	7,400,000	42,400,000	59,700,000	60,800,000	1,700,000	48,700,000	5,800,000	10,700,000	9,700,000	12,000,000

KSE-30	UBL	BAHL	NRL	HUBC	BAFL	ENGRO	MCB	FFC	SYS	CHCC	POL	LUCK	MTL	MARI	COLG
WEIGHTAGE	5.94%	4.04%	0.47%	7.72%	2.61%	6.28%	5.56%	5.83%	4.76%	1.20%	3.38%	4.52%	3.45%	4.58%	2.21%
ALLOCATION	59,400,000	40,400,000	4,700,000	77,200,000	26,100,000	62,800,000	55,600,000	58,300,000	47,600,000	12,000,000	33,800,000	45,200,000	34,500,000	45,800,000	22,100,000

The amount of 1 billion was done on the basis of these index weightages, which were obtained from PSX.

4.2.5: Call to BUY or SELL

9 =IF(AND(BV9<$GG9,BV9>0),"Purchase equivalent to limit","sell 90% of holding")

B	C	BS	BT	BU	BV	BW
		SHEL				
S.NO	DATE	NO. OF SHARES	CHANGE	Check	P/E RATIO	Action
1	1-Jan-19	28,392	-	Ok	11.28	
2	2-Jan-19	28,392	-4%	Ok	10.83	
3	3-Jan-19	28,392	-6%	Ok	10.62	
4	4-Jan-19	28,392	-7%	Ok	10.44	
5	7-Jan-19	28,392	-10%	Ok	10.19	
6	8-Jan-19	53,092	-11%	Action Required	10.06	Purchase equivalent to limit
7	9-Jan-19	53,092	-2%	Ok	9.83	
8	10-Jan-19	53,092	-2%	Ok	9.88	
9	11-Jan-19	53,092	-3%	Ok	9.81	
10	14-Jan-19	53,092	1%	Ok	10.14	
11	15-Jan-19	53,092	1%	Ok	10.17	
12	16-Jan-19	53,092	1%	Ok	10.18	
13	17-Jan-19	53,092	1%	Ok	10.19	
14	18-Jan-19	53,092	2%	Ok	10.26	
15	21-Jan-19	53,092	2%	Ok	10.30	
16	22-Jan-19	53,092	0%	Ok	10.05	
17	23-Jan-19	53,092	0%	Ok	10.09	
18	24-Jan-19	53,092	0%	Ok	10.10	
19	25-Jan-19	53,092	1%	Ok	10.14	
20	28-Jan-19	53,092	6%	Ok	10.64	
21	29-Jan-19	53,092	11%	Ok	11.17	
22	30-Jan-19	53,092	16%	Ok	11.63	
23	31-Jan-19	53,092	20%	Ok	12.04	
24	1-Feb-19	53,092	20%	Ok	12.09	

Checking by the help of P/E and average P/E ratio to sell or purchase the share, where action is required.

4.3 Portfolio Construction

For the purpose of constructing a portfolio, this study uses Rs. 1,000,000,000 as our initial investment. This amount is invested according to KSE 30 weightages provided by PSX.

Table 3: 4.3 Portfolio Construction

4.3.1 Total Cost of Investment

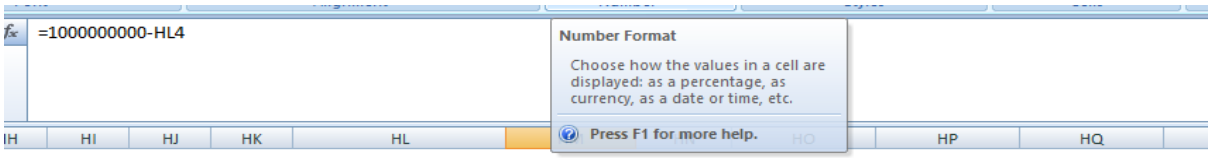
=($D$4*$E$4)+($J$4*$K$4)+($P$4*$Q$4)+($V$4*$W$4)+($AB$4*$AC$4)+($AH$4*$AI$4)+($AN$4*$AO$4)+($AT$4*$AU$4)+($AZ$4*$BA$4)+($BF$4*$BG$4)+($BL$4*$BM$4)+($BR$4*$BS$4)+($BX$4*$BY$4)+($CD$4*$CE$4)+($CJ$4*$CK$4)+($CP$4*$CQ$4)+($CV$4*$CW$4)+($DB$4*$DC$4)+($DH$4*$DI$4)+($DN$4*$DO$4)+($DT$4*$DU$4)+($DZ$4*$EA$4)+($EF$4*$EG$4)+($EL$4*$EM$4)+($ER$4*$ES$4)+($EX$4*$EY$4)+($FD$4*$FE$4)+($FJ$4*$FK$4)+($FP$4*$FQ$4)+($FV$4*$FW$4)

HI	HJ	HK	HL	HM	HN	HO	HP	HQ	HR	HS	H

Total Cost of Investment	Available Limit	Avg P/E	Market Value	Log Returns
995,500,000	4,500,000	60.85	995,500,000	
995,500,000	4,500,000	60.54	992,705,800	-0.0028
995,500,000	4,500,000	59.24	987,713,695	-0.0050
995,500,000	4,500,000	58.92	986,645,916	-0.0011
995,500,000	4,500,000	61.21	1,017,304,773	0.0306
999,999,963	37	62.90	1,041,708,710	0.0237
999,999,963	37	62.56	1,039,559,094	-0.0021
999,999,963	37	61.78	1,043,284,492	0.0036
999,999,963	37	61.17	1,040,609,833	-0.0026
999,999,963	37	62.88	1,049,029,071	0.0081
995,769,963	4,230,037	62.60	1,050,893,461	0.0018
995,769,963	4,230,037	61.66	1,040,054,623	-0.0104
995,769,963	4,230,037	62.28	1,040,637,294	0.0006
995,769,963	4,230,037	62.58	1,039,333,101	-0.0013
995,769,963	4,230,037	62.46	1,045,667,426	0.0061
995,769,963	4,230,037	64.31	1,061,748,469	0.0153
995,769,963	4,230,037	64.52	1,059,825,336	-0.0018
995,769,963	4,230,037	64.09	1,066,874,608	0.0066
991,989,963	8,010,037	65.29	1,064,805,747	-0.0019
991,989,963	8,010,037	65.89	1,067,223,022	0.0023
991,989,963	8,010,037	65.63	1,075,685,831	0.0079

Total cost of investment is calculated by multiplying number of shares with the initial prices. This work is done for 30 shares and then add all the multiplied numbers.

4.3.2: Available Limit



Total Cost of Investment	Available Limit	Avg P/E	Market Value	Log Returns
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995,500,000	4,500,000	60.54	992,705,800	-0.0028
995,500,000	4,500,000	59.24	987,713,695	-0.0050
995,500,000	4,500,000	58.92	986,645,916	-0.0011
995,500,000	4,500,000	61.21	1,017,304,773	0.0306
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999,999,963	37	61.78	1,043,284,492	0.0036
999,999,963	37	61.17	1,040,609,833	-0.0026
999,999,963	37	62.88	1,049,029,071	0.0081
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995,769,963	4,230,037	62.28	1,040,637,294	0.0006
995,769,963	4,230,037	62.58	1,039,333,101	-0.0013
995,769,963	4,230,037	62.46	1,045,667,426	0.0061
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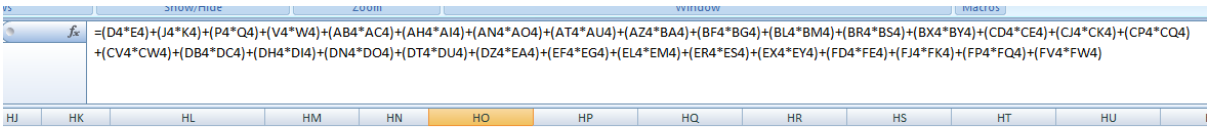
995,769,963	4,230,037	64.52	1,059,825,336	-0.0018
995,769,963	4,230,037	64.09	1,066,874,608	0.0066
991,989,963	8,010,037	65.29	1,064,805,747	-0.0019
991,989,963	8,010,037	65.89	1,067,223,022	0.0023
991,989,963	8,010,037	65.63	1,075,685,831	0.0079

Available limit is derived by subtracting total cost of investing from hypothetical portfolio amount.

4.3.3: Market Value



DATE	Total Cost of Investment	Available Limit	Avg P/E	Market Value	Log Returns
1-Jan-19	995,500,000	4,500,000	60.85	995,500,000	
2-Jan-19	995,500,000	4,500,000	60.54	992,705,800	-0.0028
3-Jan-19	995,500,000	4,500,000	59.24	987,713,695	-0.0050
4-Jan-19	995,500,000	4,500,000	58.92	986,645,916	-0.0011
7-Jan-19	995,500,000	4,500,000	61.21	1,017,304,773	0.0306
8-Jan-19	999,999,963	37	62.90	1,041,708,710	0.0237
9-Jan-19	999,999,963	37	62.56	1,039,559,094	-0.0021
10-Jan-19	999,999,963	37	61.78	1,043,284,492	0.0036
11-Jan-19	999,999,963	37	61.17	1,040,609,833	-0.0026
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15-Jan-19	995,769,963	4,230,037	62.60	1,050,893,461	0.0018
16-Jan-19	995,769,963	4,230,037	61.66	1,040,054,623	-0.0104
17-Jan-19	995,769,963	4,230,037	62.28	1,040,637,294	0.0006
18-Jan-19	995,769,963	4,230,037	62.58	1,039,333,101	-0.0013
21-Jan-19	995,769,963	4,230,037	62.46	1,045,667,426	0.0061
22-Jan-19	995,769,963	4,230,037	64.31	1,061,748,469	0.0153
23-Jan-19	995,769,963	4,230,037	64.52	1,059,825,336	-0.0018
24-Jan-19	995,769,963	4,230,037	64.09	1,066,874,608	0.0066
25-Jan-19	991,989,963	8,010,037	65.29	1,064,805,747	-0.0019
28-Jan-19	991,989,963	8,010,037	65.89	1,067,223,022	0.0023

Market Value is calculated by multiplying number of shares with the share price.

Table 4.3.4: Log Returns

=LN(HO5/HO4)

HJ	HK	HL	HM	HN	HO	HP	HQ

DATE	Total Cost of Investment	Available Limit	Avg P/E	Market Value	Log Returns
1-Jan-19	995,500,000	4,500,000	60.85	995,500,000	
2-Jan-19	995,500,000	4,500,000	60.54	992,705,800	-0.0028
3-Jan-19	995,500,000	4,500,000	59.24	987,713,695	-0.0050
4-Jan-19	995,500,000	4,500,000	58.92	986,645,916	-0.0011
7-Jan-19	995,500,000	4,500,000	61.21	1,017,304,773	0.0306
8-Jan-19	999,999,963	37	62.90	1,041,708,710	0.0237
9-Jan-19	999,999,963	37	62.56	1,039,559,094	-0.0021
10-Jan-19	999,999,963	37	61.78	1,043,284,492	0.0036
11-Jan-19	999,999,963	37	61.17	1,040,609,833	-0.0026
14-Jan-19	999,999,963	37	62.88	1,049,029,071	0.0081
15-Jan-19	995,769,963	4,230,037	62.60	1,050,893,461	0.0018
16-Jan-19	995,769,963	4,230,037	61.66	1,040,054,623	-0.0104
17-Jan-19	995,769,963	4,230,037	62.28	1,040,637,294	0.0006
18-Jan-19	995,769,963	4,230,037	62.58	1,039,333,101	-0.0013
21-Jan-19	995,769,963	4,230,037	62.46	1,045,667,426	0.0061
22-Jan-19	995,769,963	4,230,037	64.31	1,061,748,469	0.0153
23-Jan-19	995,769,963	4,230,037	64.52	1,059,825,336	-0.0018
24-Jan-19	995,769,963	4,230,037	64.09	1,066,874,608	0.0066
25-Jan-19	991,989,963	8,010,037	65.29	1,064,805,747	-0.0019
28-Jan-19	991,989,963	8,010,037	65.89	1,067,223,022	0.0023
29-Jan-19	991,989,963	8,010,037	65.63	1,075,685,831	0.0079

Here the log return is derived by the help of log difference.

4.4 Stationarity Test/ Unit Root Test

After constructing the portfolio, this study determine its market value and then calculate the log differences to analyze the returns. Following this, this study assess the stationarity of these returns using the Augmented Dickey-Fuller (ADF) test in EViews Software. This step is essential to confirm that the time series data is stationary, as stationarity is a fundamental requirement for many econometric and financial modeling approaches. Ensuring stationarity allows us to proceed with subsequent analyses and accurately model the portfolio's behavior.

Table 4: 4.4 Stationarity Test/ Unit Root Test

Unit Root with Break Test on LOG_RETURNS

Null Hypothesis: LOG_RETURNS has a unit root
Trend Specification: Intercept only
Break Specification: Intercept only
Break Type: Innovational outlier

Break Date: 25
Break Selection: Minimize Dickey-Fuller t-statistic
Lag Length: 0 (Automatic - based on Schwarz information criterion, maxlag=22)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-67.33898	< 0.01
Test critical values:	1% level	-4.949133	
	5% level	-4.443649	
	10% level	-4.193627	

The Augmented Dickey-Fuller (ADF) test was conducted to determine if the time series is stationary or not. The test results showed a probability value less than 0.05, leading to the rejection of the null hypothesis that the time series is non-stationary. Therefore, we conclude that the time series is stationary.

4.5 Calculation of Volatility & VaR

This study calculated the volatility of each company's log returns over a five-year period using Excel. This involved determining the standard deviation of the log returns.

Using the historical volatility calculated in the previous step, this study estimated the Value at Risk (VaR) for each company employing the Variance-Covariance method. This method assumes normally distributed returns and uses the standard deviation to estimate potential losses.

Various GARCH models is applied to the market log returns to estimate volatility. This study estimates VaR for each company using the volatility estimated from the GARCH models. This step involved incorporating the GARCH-derived volatility into the Variance-Covariance method.

Finally, compared VaR values obtained from both the variance covariance volatility and GARCH-derived volatility with the actual observed losses for each company. This comparison enabled us to assess the accuracy and reliability of each method in predicting potential market losses.

Table 5: 4.5 Calculation of Volatility & VaR

4.5.1 Calculation of VaR by Time Invariant Method

Date	Share Price	Logarithmic Returns	1 Day VaR	Actual Loss
1-Jan-19	154.99			
2-Jan-19	154.16	-0.005369575	8.67	0.83
3-Jan-19	149.96	-0.027622432	8.43	4.20
4-Jan-19	150.08	0.000799893	8.44	(0.12)
7-Jan-19	156.69	0.043100846	8.81	(6.61)
8-Jan-19	164.19	0.046754963	9.23	(7.50)
9-Jan-19	165.71	0.009214979	9.32	(1.52)

10-Jan-19	171.16	0.032359519	9.62	(5.45)
11-Jan-19	171.88	0.004197767	9.66	(0.72)
14-Jan-19	174.66	0.016044668	9.82	(2.78)
15-Jan-19	183.14	0.047409661	10.30	(8.48)
16-Jan-19	180.17	-0.016350038	10.13	2.97
17-Jan-19	178.83	-0.007465216	10.05	1.34
18-Jan-19	178.05	-0.004371224	10.01	0.78
21-Jan-19	182.74	0.025999968	10.27	(4.69)
22-Jan-19	183.81	0.005838238	10.33	(1.07)
23-Jan-19	182.90	-0.00496306	10.28	0.91
24-Jan-19	185.78	0.015623623	10.45	(2.88)
25-Jan-19	184.66	-0.006046882	10.38	1.12
28-Jan-19	185.05	0.002109762	10.40	(0.39)
29-Jan-19	188.68	0.0194264	10.61	(3.63)
30-Jan-19	189.44	0.004019893	10.65	(0.76)

This study estimated the Value at Risk (VaR) for each company employing the Variance-Covariance Time invariant method.

4.5.2 Calculation of Volatility by different GARCH type Models

MODEL NAME	Omega	Alpha1	Alpha2	Alpha3	Beta1	Beta2	Beta3	Log Likelihood	AIC
Garch(1,0)	0.0003164***	0.4113***	0	0	0	0	0	3063.226	-4.943822
Garch(2,0)	0.0002732***	0.4084***	0.1362**	0	0	0	0	3072.879	-4.957801
Garch(3,0)	0.0002348***	0.2108**	0.1111**	0.3379***	0	0	0	3079.214	-4.96642
Garch(1,1)	1.377e-05 ***	1.035e-01***	0	0	0.8814***	0	0	3113.447	-5.02334
Garch(1,2)	0.00002677***	0.1868***	0	0	0.08731**	0.6952***	0	3120.269	-5.032744
Garch(2,1)	0.00001317***	1.00E-08	0.114***	0	0.875***	0	0	3117.764	-5.028697
Garch(2,2)	0.00002649***	0.1774***	1.83E-02	0	0.07009*	0.7058***	0	3120.773	-5.031944
Garch(3,1)	0.00001382***	1.00E-08	0.05628.	0.06793***	0	0	0	3119.163	-5.029343
Garch(1,3)	0.0000269***	0.1865***	0	0	8.72E-02	0.6952***	1.00E-08	3119.961	-5.030631
Garch(2,3)	0.00003412***	9.72E-03	0.1875***	0	1.00E-08	1.00E-08	0.7428***	3155.417	-5.086296
Garch(3,2)	0.00002523***	1.00E-08	0.03347.	0.2022***	0.09325**	0.6551***	0	3131.107	-5.047022
Garch(3,3)	0.0000277***	1.00E-08	0.03837***	0.218***	1.00E-08	0.6282***	9.67E-02	3133.362	-5.04905

Various GARCH models were applied to the market log returns to estimate volatility. These values are obtained by running program in R.

4.5.3 Calculation of VaR by using different GARCH type Models

Date	Share Price	GARCH (2,3)	1 Day VaR - GARCH (2,3)
1-Jan-19	134.07		
2-Jan-19	134.5	-0.000022	1.457
3-Jan-19	129.98	-0.000022	1.408
4-Jan-19	130.18	-0.000022	1.410
7-Jan-19	135.1	-0.000022	1.464
8-Jan-19	139.14	-0.000022	1.507
9-Jan-19	139.62	-0.000022	1.512
10-Jan-19	142.96	-0.000022	1.549
11-Jan-19	142.4	-0.000022	1.543
14-Jan-19	142.06	-0.000022	1.539
15-Jan-19	146.23	-0.000022	1.584
16-Jan-19	144.91	-0.000022	1.570
17-Jan-19	144.54	-0.000022	1.566
18-Jan-19	145.15	-0.000022	1.572
21-Jan-19	147.87	-0.000022	1.602
22-Jan-19	148.62	-0.000022	1.610
23-Jan-19	149.47	-0.000022	1.619
24-Jan-19	149.62	-0.000022	1.621
25-Jan-19	148.26	-0.000022	1.606
28-Jan-19	147.59	-0.000022	1.599
29-Jan-19	150	-0.000022	1.625
30-Jan-19	150.2	-0.000022	1.627

This study calculated the VaR for each company using the volatility estimated from the GARCH models.

4.5.4Comparing calculated VaR(s) by Actual loss

Diff B/w Loss and VaR	Chk - Time Invariant	Diff B/w Loss and GARCH (2,3) VaR	Chk - GARCH (2,3) VaR
7.078	Ok	1.887	Ok
1.905	Ok	-3.112	Breached
6.635	Ok	1.610	Ok
11.598	Ok	6.384	Ok
10.918	Ok	5.547	Ok
7.382	Ok	1.992	Ok
10.407	Ok	4.889	Ok
6.479	Ok	0.983	Ok
6.682	Ok	1.199	Ok
11.398	Ok	5.754	Ok

5.843	Ok	0.250	Ok
6.775	Ok	1.196	Ok
7.785	Ok	2.182	Ok
10.029	Ok	4.322	Ok
8.096	Ok	2.360	Ok
8.238	Ok	2.469	Ok
7.546	Ok	1.771	Ok
5.969	Ok	0.246	Ok
6.625	Ok	0.929	Ok
9.825	Ok	4.035	Ok
7.625	Ok	1.827	Ok

This study compared the VaR values obtained from both the historical volatility and GARCH-derived volatility with the actual observed losses for each company.

CHAPTER 05

EMPERICAL RESULTS

Accompanying the extensive construction of a hypothetical KSE-30 index portfolio, this study conclude at these insights.

5.1 Descriptive Analysis

Mean of Market Value	2,172,379,961
Mean of Log return	0.001
Median of Market Value	2,150,247,216
Median of Log return	0.00004
Variance of Market Value	6.608E+17
Variance of Log return	0.00044
Standard Deviation of Market Value	3E+17
Standard Deviation of Log return	2E-04
Skewness of Market Value	0.3804
Skewness of Log return	2.3627
Kurtosis of Market Value	-0.3236
Kurtosis of Log return	27.0160
Q1 of Market Value	1,518,988,646
Q1 of Log return	-0.0092
Q3 of Market Value	2,706,196,652
Q3 of Log return	0.0105
IQR of Market value	1,187,208,006
IQR of Log return	0.01974

The descriptive evaluation of the log variance and the market value is presented below. Throughout the preceding five years, the average market value of the portfolio's 30 companies stands at Rs. 2,172,379,961.

The value of Rs. 2,150,247,216 indicates that during the five years preceding the present, the market values of half of the companies in the portfolio were higher than Rs. 2,150,247,216, whereas the market values of the other half were less than this amount.

Since the mean exceeds the median, the distribution of market values is skewed to the right. This suggests that the portfolio comprises a wider range of companies with lower market values than with higher market values. The right-hand tail also indicates that just a few percent of companies may yield returns that are noticeably higher than the average.

6.608E+17 indicates a significant deviation of market prices from the mean. A greater variance indicates greater dispersion. The significant variance here validates our previous finding regarding the portfolio's market value dispersion.

3.304E+17 is the square root of variance, which serves as a different approach of measuring the extent to which the data deviates from the mean. The market values tend to deviate significantly from the average market value, reflected by the large standard deviation figure.

The symmetry of the market value distribution is determined at 0.38004. The subsequent distribution, which has a positive skew, has a "tail" that points to the right and validates our previous assumption that many companies have lower market values.

This measuring parameter defines a distribution's tails. It concerns the present value as well as the kurtosis of the log return in this instance. A kurtosis of 3 indicates a distribution with wider tails than a normal distribution, while a kurtosis of less than three also shows wider tails. The market value distribution has narrower tails than a normal distribution, which is demonstrated by the market value's kurtosis of -0.32, a value less than 3. However, as indicated by the log return's kurtosis of 27.02, which is significantly greater than a normal distribution, the resulting distribution of log returns has significantly larger tails than a normal distribution.

The data is split into four equal quarters by these values. The median of the lowest part of the data is indicated by the first quartile (Q1), the median of the entire data obtained from the second quartile (Q2), and the median of the upper half of the data set by the third quartile (Q3) The difference between the first quartile (Q1) and the third quartile (Q3) is referred as the interquartile range (IQR). The market value and log return Q1, Q3, and IQR are shown in the table.

The market value for Q1 is 1,518,988,646, while Q3 is 2,706,196,652. It implies that the middle 50% of companies' market values vary between 1.52 billion and 2.71 billion. The IQR (1,187,208,006) indicates the variance in the middle 50% of the data.

The log return's Q1 is -0.0092, while Q3 is 0.0105. The meaning of "log return" relates to the natural logarithm of the ratio between consecutive stock prices. Positive log returns represent price increases, whereas negative log returns represent price decreases.

The fact that the Q1 is negative suggests that the median log return over this period was negative, which means that the median company's stock price went down over this period. The IQR is 0.0197, which signifies the spread of the middle 50% of the log return data.

5.2 Encompass Details Regarding Volatility Estimation

COMPANY	MODEL NAME	Omega	Alpha1	Alpha2	Alpha3	Beta1	Beta2	Beta3	Log Likelihood	AIC
OGDC	GARCH (2,3)	0.00003143*	0.1171*	7.94E-02		1.00E-08	1.00E-08	0.7409*	3097.437	-4.992629
PRL	GARCH (1,2)	7.49E-04	0.2763***			0.2377***	1.00E-08		2350.268	-3.788802
TRG	GARCH (1,2)	0.00004321*	0.152***			3.68E-01	0.4477.		2480.292	-3.998856
PSO	GARCH (2,3)	0.00007895.	0.09526***	2.94E-03		1.00E-08	1.35E-01	0.6273***	2886.514	-4.65188
SEARL	GARCH (2,3)	8.94E-04	1.00E-08	0.08194***		1.00E-08	1.00E-08	1.00E-08	2552.037	-4.11153
HBL	GARCH (1,2)	0.0001359***	0.6863***			0.145**	1.11E-02		3226.014	-5.203576
EFERT	GARCH (1,2)	0.00005699***	0.2014***			5.81E-01	1.00E-08		3455.687	-5.574615
MEBL	GARCH (2,3)	0.00005873**	0.3833***	1.00E-08		1.00E-08	1.99E-03	0.59***	3116.872	-5.024025
TPLP	GARCH (1,2)	0.0000484**	0.1369***			0.4669**	0.3738*		2327.615	-3.752205
PPL	GARCH (2,3)	0.00004989***	0.1389***	6.88E-03		1.00E-08	0.1562**	6.20E-01	2910.313	-4.690328
SHEL	GARCH (1,2)	0.00008036***	0.3391***			1.00E-08	0.6151***		2804.467	-4.522564
ATRL	GARCH (2,3)	0.000104*	0.1023***	0.1246**		2.35E-01	1.00E-08	4.370e-01.	2580.135	-4.156923
DGKC	GARCH (1,2)	0.00004371*	0.09921**			4.10E-01	4.28E-01		2795.371	-4.507869
SNGP	GARCH (2,3)	0.0001616*	0.217**	1.00E-08		2.76E-01	1.00E-08	2.93E-01	2752.731	-4.435752
UBL	GARCH (2,3)	0.00006029***	0.1616***	2.02E-02		3.65E-01	1.00E-08	0.2792*	3238.755	-5.22093

BAHL	GARCH (1,2)	0.0000625***	0.1519***			0.6289**	3.99E-02		3261.962	-5.261651
NRL	GARCH (1,2)	0.00007003**	0.1536***			0.5556**	2.22E-01		2599.572	-4.191554
HUBC	GARCH (1,2)	0.00005166***	0.1206**			0.6031.	1.60E-01		3100.118	-5.000191
BAFL	GARCH (1,2)	0.00003275**	0.1215**			5.61E-01	2.29E-01		3220.974	-5.195434
ENGRO	GARCH (2,3)	0.00004066*	0.1406*	0.1849*		1.00E-08	0.3879.	1.35E-01	3493.871	-5.633072
MCB	GARCH (1,2)	0.00003436**	0.1542***			9.22E-02	6.136e-01***		3462.832	-5.586158
FFC	GARCH (1,2)	0.00001385*	0.1685*			7.48E-01	1.00E-08		3776.343	-6.092638
SYS	GARCH(3,3)	0.0000165**	1.00E-08	0.2657***	1.00E-08	1.00E-08	8.11E-01	1.96E-03	2754.334	-4.436726
CHCC	GARCH (1,2)	0.0000547**	0.1111**			0.8188*	1.00E-08		2726.776	-4.397054
POL	GARCH (1,2)	0.00008578***	0.2503***			0.4821***	1.00E-08		3340.514	-5.388553
LUCK	GARCH (1,2)	0.00004291***	0.1501***			1.99E-01	0.5473***		3129.319	-5.047365
MTL	GARCH (1,2)	4.11E-05	1.00E-08			1.08E-03	0.969***		2342.994	-3.777705
MARI	GARCH (2,3)	0.00003973***	0.1879***	0.1196.		1.79E-02	5.60E-02	0.4726***	3461.792	-5.581247
COLG	GARCH (1,2)	0.0003113***	0.3513***			0.4747***	1.00E-08		2585.114	-4.168197

In order to determine which GARCH model best captured the data's volatility, this review evaluated numerous of models.
The factors analyzed are as follows:

Persistence: The manner in which that previous volatility (shocks) affects future volatility is captured by GARCH models. This persistence is reflected in the model's sum of alpha (α) and beta (β) if equal to or less than 1 implies that past volatility shocks have a significant impact on future volatility.

Akaike Information Criterion, or AIC: It's a goodness-of-fit metric that penalizes complexity. A reduced AIC value indicates a more favorable compromise between upholding a comparatively simplistic model and accurately fitting the data.

Log-likelihood: This measure shows how well the model explains the data that have been observed. A greater log-likelihood number indicates a higher likelihood that the observed data has been generated by the model.

The GARCH model that best reflects the volatility dynamics while retaining a reasonable degree of complexity and avoiding overfitting can be found by combining the evaluations of the three components as shown below.

5.3 Estimation of VaR

COMPANY	Avg. Distance b/w Loss and Time Invariant VaR	Avg. Distance b/w Loss and GARCH (2,3) VaR	No. of breaches in Avg. Distance b/w Loss and Time Invariant VaR	No. of breaches in Avg. Distance b/w Loss and GARCH (2,3) VaR
OGDC	5.0518	1.0934	20	314
PRL	1.6578	1.8545	3	2
TRG	6.4800	2.0048	1	241
PSO	9.9772	7.2165	21	54
SEARL	12.206	2.1664	13	0
HBL	5.5139	7.6688	18	13
EFERT	2.7449	3.1752	32	22
MEBL	5.1259	2.9378	20	49
TPLP	1.4714	0.9104	21	91
PPL	5.3216	0.0614	21	0
SHEL	9.8603	8.0176	13	23
ATRL	11.8312	8.7866	27	56
DGKC	4.6760	1.237	21	281
SNGP	3.2214	2.4888	24	56
UBL	5.7690	4.5232	24	37
BAHL	2.8085	0.4933	18	0
NRL	17.0089	13.9855	23	50
HUBC	3.7773	3.6067	20	21
BAFL	1.6669	1.2773	19	34
ENGRO	10.6196	6.5349	24	66
MCB	5.8275	4.9110	25	34
FFC	3.1665	1.9356	24	69
SYS	24.9415	35.5266	10	1
CHCC	7.2840	5.6564	20	49
POL	16.3154	5.1851	27	177
LUCK	27.3571	33.4557	23	12

MTL	58.9723494	9.29178483	8	159
MARI	58.13797228	48.52809891	22	40
COLG	159.5957336	86.75059305	13	72

It is most likely the case that the mean distance between the loss and the GARCH (2, 3) VaR is less than that of the time-invariant VaR at average. When compared to the time-invariant VaR, a smaller average distance indicates that the GARCH (2, 3) model yields a more accurate estimate of VaR.

Based on the findings, it can be concluded that the GARCH (2, 3) model yields more accurate and precise VaR estimates overall and represents the time-varying character of volatility better than the time-invariant VaR model as shown below.

CHAPTER 06

CONCLUSION

Value-at-Risk is a widely acclaimed method for measuring the risk worldwide. Due to a number of regulatory obligations, various approaches of VaR has been practiced by FIs operating in Pakistan. So far, the VaR is mostly measured and reported by the risk controllers in FIs, whereas, the risk takers by and large, are not using VaR as a tool for decision making. In order to utilize this risk metric for decision making purposes various approaches of VaR are yet to be explored in the context of Pakistani financial markets which are already applied in developed financial markets.

This study is an attempt to fill this gap, where we explored an alternative approach of VaR on a portfolio comprising equity securities enlisted in PSX. For this purpose, we have collected the daily data for the constituents of KSE 30 index from January 2019 to December 2023. With these securities, we have constructed a PKR 1 Bn hypothetical portfolio. While constructing the hypothetical portfolio, we tried to mirror the considerations of portfolio managers. For instance, if the market value of any security falls below 10% of the original cost of investment, it may prompt equity managers to revise their investment position, by means of either selling the security or acquire more shares of it for reducing the cost of investment.

We based the criterion chooses for bifurcating the above underlined binary by means of widely celebrated Price-to-Earning (P/E) ratio. Once the portfolio mix is constructed, we estimated it's volatility by means of various GARCH type models as proposed by Bollerslev (1986). Amongst numerous applied models, we found GARCH (2, 3) and GARCH (1, 2) are found to be most appropriate for capturing the volatility of this hypothetically constructed portfolio. In the later stage, we estimated VaR by means of variance-covariance approach for the components of our portfolio on standalone basis.

While going for the application of VaR estimation via variance-covariance approach we used the conditional heteroskedastic approach for accounting of the variance factor. Thus, we estimated the dynamic variance, instead of conventionally deployed static variance approaches. Finally, we have compared the VaR estimation came from dynamic approach with the static approach. Our findings proven that VaR estimated through dynamic variance is a more effective tool for quantifying risk as compared to the VaR estimated through keeping variance constant. The prime qualifiers of the previous statements are (a) it is more closer to the actual losses which were historically incurred and (b) it creates more breaches for the VaR limit, which implies more action points for the portfolio managers.

RECOMMENDATIONS

While researching for more than a year about this project, this study noticed various pain points and gaps that can be filled if due measures are taken by the authorities and people running the capital markets.

Pakistan's stock market largely runs on human sentiments. This has been the case for almost since the exchange came into being. Normally, there exists a causal relation between the stock markets and the economy of the country, but unfortunately this is not the case with our stock market.

The main reason behind this is the lack of necessary knowledge about investment ethics and healthy routines that are widely adopted among the western stock markets, which also include transparent and unbiased analysis and research, which is imperative to earn better gains and mitigate any risks and losses. Since many companies listed on PSX do not share transparent and authentic financial statements, it becomes a hurdle for analyzing the value of the company so that people could invest in healthy companies.

According to this study, these models should help the research analysts and equity managers to guide their investors in a better way and to inform them about the associated risks, all this with better accuracy and less biasness. This study believe that the equity traders and other related personnel should adopt these models in their research as it could help increase healthy investment practices without too much risks and biased buying, followed by acting upon the herd mentality, thereby appreciating a stock well above its fair value.

REFERENCES

1. Umar, Kumail, Mehreen (2023). Asymmetric volatility structure of equity returns: Evidence from an emerging market.
2. Kingsley Arum (2019). Volatility Modelling using Arch and Garch Models (A Case Study of the Nigerian Stock Exchange)
3. Yan Chin Liu and Ming Li (2006). A calculation of VaR based on GARCH model.
4. Lim and Kun Sek (2013). Comparing the performances of GARCH-type models in capturing the stock market volatility in Malaysia.
5. Antin Sorin Gabriel (2012). Evaluating the Forecasting Performance of GARCH Models. Evidence from Romania
6. Timotheos, alexandros and Stavros (2004). The use of GARCH models in VaR estimation
7. E. Mao, L. Xu, and W. Tian (2012). A Dynamic Computing Research for Value at Risk (VaR) of Shanghai Stock Market Based on the GARCH Model
8. Hashmi, Hassan, Asim, Zubair, Muhammad (2022). The dynamics of Industrial Level Volatility Spillovers in Emerging Equity Market: Evidence from PSX
9. Aleksandra and Magdalena Joanna (2021). Model Risk of VaR under Different Approaches: Study on European Banks
10. Sang Jin LeEKorean (2008). Model Selection for Estimating Portfolio VAR in Korean Stock Market
11. Saint Kuttu (2014). Return and volatility dynamics among four African equity markets: A multivariate VAR-EGARCH analysis

APPENDIX

01: Summary of GARCH (1, 0) on log return of Market Value

```
> summary(f1)

Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(1, 0), data = x)

Mean and Variance Equation:
 data ~ garch(1, 0)
<environment: 0x000001d3edeaac18>
 [data = x]

Conditional Distribution:
 norm

Coefficient(s):
        mu       omega      alpha1
0.00022519  0.00031642  0.41133683

Std. Errors:
 based on Hessian

Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu      2.252e-04   5.739e-04    0.392    0.695
omega   3.164e-04   2.257e-05   14.018  < 2e-16 ***
alpha1  4.113e-01   1.036e-01    3.969 7.23e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 3063.226    normalized:  2.474334
```

The main findings obtained from this summary are alpha and log likelihood.

02: Summary of GARCH (2, 0) on log return of Market Value

```
> summary(f2)

Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(2, 0), data = x)

Mean and Variance Equation:
 data ~ garch(2, 0)
<environment: 0x000001d3eefe7660>
 [data = x]

Conditional Distribution:
 norm

Coefficient(s):
        mu       omega      alpha1      alpha2
0.00015165  0.00027318  0.40844660  0.13615574

Std. Errors:
 based on Hessian

Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu      1.516e-04   5.415e-04    0.280  0.77943
omega   2.732e-04   2.194e-05   12.451  < 2e-16 ***
alpha1  4.084e-01   9.960e-02    4.101 4.12e-05 ***
alpha2  1.362e-01   4.200e-02    3.242  0.00119 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 3072.879    normalized:  2.482131
```

The main findings obtained from this summary are alpha1, alpha2 and log likelihood.

03: Summary of GARCH (3, 0) on log return of Market Value

```
Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(3, 0), data = x)

Mean and Variance Equation:
 data ~ garch(3, 0)
<environment: 0x000001d3e751efa0>
 [data = x]

Conditional Distribution:
 norm

Coefficient(s):
        mu       omega      alpha1      alpha2      alpha3
0.00162280  0.00023485  0.21078388  0.11114131  0.33787500

Std. Errors:
 based on Hessian

Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu     1.623e-03   5.923e-04    2.740 0.006152 **
omega  2.348e-04   2.176e-05   10.792  < 2e-16 ***
alpha1 2.108e-01   6.491e-02    3.248 0.001164 **
alpha2 1.111e-01   3.568e-02    3.115 0.001838 **
alpha3 3.379e-01   1.007e-01    3.357 0.000789 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 3079.214    normalized:  2.487249
```

The main findings obtained from this summary are alpha1, alpha2, alpha3 and log likelihood.

04: Summary of GARCH (1, 1) on log return of Market Value

```
Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(1, 1), data = x)

Mean and Variance Equation:
 data ~ garch(1, 1)
<environment: 0x000001d3eb99b910>
 [data = x]

Conditional Distribution:
 norm

Coefficient(s):
        mu       omega      alpha1       beta1
1.6989e-03  1.3775e-05  1.0349e-01  8.8136e-01

Std. Errors:
 based on Hessian

Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu     1.699e-03   5.109e-04    3.325 0.000883 ***
omega  1.377e-05   3.053e-06    4.511 6.44e-06 ***
alpha1 1.035e-01   1.597e-02    6.481 9.13e-11 ***
beta1  8.814e-01   1.247e-02   70.664  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 3113.447    normalized:  2.514901
```

05: Summary of GARCH (1, 2) on log return of Market Value

```
Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(1, 2), data = x)

Mean and Variance Equation:
 data ~ garch(1, 2)
<environment: 0x000001d3ebae3520>
 [data = x]

Conditional Distribution:
 norm

Coefficient(s):
         mu        omega       alpha1        beta1        beta2
0.00139828  0.00002677  0.18681897  0.08731091  0.69515098

Std. Errors:
 based on Hessian

Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu      1.398e-03   5.099e-04    2.742  0.00610 **
omega   2.677e-05   5.788e-06    4.625 3.74e-06 ***
alpha1  1.868e-01   2.866e-02    6.519 7.07e-11 ***
beta1   8.731e-02   2.980e-02    2.930  0.00339 **
beta2   6.952e-01   3.132e-02   22.193  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 3120.269    normalized:  2.520411
```

The main findings obtained from this summary are alpha1, beta1, beta2 and log likelihood.

06: Summary of GARCH (2, 1) on log return of Market Value

```
Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(2, 1), data = x)

Mean and Variance Equation:
 data ~ garch(2, 1)
<environment: 0x000001d3e73dd9a8>
 [data = x]

Conditional Distribution:
 norm

Coefficient(s):
         mu        omega       alpha1       alpha2        beta1
1.9091e-03  1.3175e-05  1.0000e-08  1.1398e-01  8.7503e-01

Std. Errors:
 based on Hessian

Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu      1.909e-03   4.975e-04    3.837 0.000124 ***
omega   1.317e-05   2.992e-06    4.404 1.06e-05 ***
alpha1  1.000e-08   1.704e-02    0.000 1.000000
alpha2  1.140e-01   2.378e-02    4.794 1.63e-06 ***
beta1   8.750e-01   1.231e-02   71.080  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 3117.764    normalized:  2.518387
```

The main findings obtained from this summary are alpha1, alpha2, beta1 and log likelihood.

07: Summary of GARCH (2, 2) on log return of Market Value

```
Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(2, 2), data = x)

Mean and Variance Equation:
 data ~ garch(2, 2)
<environment: 0x000001d3e7d49168>
 [data = x]

Conditional Distribution:
 norm

Coefficient(s):
         mu        omega       alpha1       alpha2        beta1        beta2
1.4421e-03   2.6493e-05   1.7738e-01   1.8272e-02   7.0094e-02   7.0576e-01

Std. Errors:
 based on Hessian

Error Analysis:
         Estimate  Std. Error  t value Pr(>|t|)
mu      1.442e-03   5.115e-04    2.820  0.00481 **
omega   2.649e-05   5.859e-06    4.522 6.13e-06 ***
alpha1  1.774e-01   3.010e-02    5.893 3.80e-09 ***
alpha2  1.827e-02   1.881e-02    0.971  0.33131
beta1   7.009e-02   3.232e-02    2.169  0.03012 *
beta2   7.058e-01   3.167e-02   22.284  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 3120.773    normalized:  2.520818
```

The main findings obtained from this summary are alpha1, alpha2, beta1, beta2 and log likelihood.

08: Summary of GARCH (3, 1) on log return of Market Value

```
Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(3, 1), data = x)

Mean and Variance Equation:
 data ~ garch(3, 1)
<environment: 0x000001d3eb958f60>
 [data = x]

Conditional Distribution:
 norm

Coefficient(s):
         mu        omega       alpha1       alpha2       alpha3        beta1
1.8702e-03   1.3818e-05   1.0000e-08   5.6282e-02   6.7927e-02   8.6555e-01

Std. Errors:
 based on Hessian

Error Analysis:
         Estimate  Std. Error  t value Pr(>|t|)
mu      1.870e-03   4.931e-04    3.792 0.000149 ***
omega   1.382e-05   3.186e-06    4.337 1.45e-05 ***
alpha1  1.000e-08   1.657e-02    0.000 1.000000
alpha2  5.628e-02   3.371e-02    1.670 0.094961 .
alpha3  6.793e-02   3.463e-02    1.962 0.049804 *
beta1   8.656e-01   1.416e-02   61.113  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 3119.163    normalized:  2.519518
```

The main findings obtained from this summary are alpha1, alpha2, alpha3, beta1 and log likelihood.

09: Summary of GARCH (1, 3) on log return of Market Value

```
Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(1, 3), data = x)

Mean and Variance Equation:
 data ~ garch(1, 3)
<environment: 0x000001d3ebba3518>
 [data = x]

Conditional Distribution:
 norm

Coefficient(s):
        mu       omega      alpha1       beta1       beta2       beta3
1.3997e-03  2.6896e-05  1.8653e-01  8.7208e-02  6.9515e-01  1.0000e-08

Std. Errors:
 based on Hessian

Error Analysis:
         Estimate  Std. Error  t value Pr(>|t|)
mu      1.400e-03   5.116e-04    2.736  0.00622 **
omega   2.690e-05   6.404e-06    4.200 2.67e-05 ***
alpha1  1.865e-01   2.926e-02    6.376 1.82e-10 ***
beta1   8.721e-02   8.320e-02    1.048  0.29453
beta2   6.952e-01   3.139e-02   22.144  < 2e-16 ***
beta3   1.000e-08   6.772e-02    0.000  1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 3119.961    normalized:  2.520162
```

The main findings obtained from this summary are alpha1, beta1, beta2, beta3 and log likelihood.

10: Summary of GARCH (1, 3) on log return of Market Value

```
Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(2, 3), data = x)

Mean and Variance Equation:
 data ~ garch(2, 3)
<environment: 0x000001d3eda873b0>
 [data = x]

Conditional Distribution:
 norm

Coefficient(s):
        mu       omega      alpha1      alpha2       beta1       beta2       beta3
1.5062e-03  3.4122e-05  9.7147e-03  1.8752e-01  1.0000e-08  1.0000e-08  7.4281e-01

Std. Errors:
 based on Hessian

Error Analysis:
         Estimate  Std. Error  t value Pr(>|t|)
mu      1.506e-03   4.911e-04    3.067  0.00216 **
omega   3.412e-05   6.118e-06    5.578 2.44e-08 ***
alpha1  9.715e-03   1.662e-02    0.585  0.55887
alpha2  1.875e-01   2.589e-02    7.243 4.39e-13 ***
beta1   1.000e-08   1.909e-02    0.000  1.00000
beta2   1.000e-08   2.194e-02    0.000  1.00000
beta3   7.428e-01   2.726e-02   27.253  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 3155.417    normalized:  2.548802
```

The main findings obtained from this summary are alpha1, alpha2, beta1, beta2, beta3 and log likelihood.

11: Summary of GARCH (3, 2) on log return of Market Value

```
Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(3, 2), data = x)

Mean and Variance Equation:
 data ~ garch(3, 2)
<environment: 0x000001d3ebfd1208>
 [data = x]

Conditional Distribution:
 norm

Coefficient(s):
         mu       omega      alpha1      alpha2      alpha3       beta1       beta2
1.6563e-03  2.5235e-05  1.0000e-08  3.3470e-02  2.0224e-01  9.3254e-02  6.5505e-01

Std. Errors:
 based on Hessian

Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu      1.656e-03   4.845e-04    3.418  0.00063 ***
omega   2.523e-05   5.749e-06    4.390 1.14e-05 ***
alpha1  1.000e-08   1.320e-02    0.000  1.00000
alpha2  3.347e-02   1.781e-02    1.880  0.06017 .
alpha3  2.022e-01   3.436e-02    5.886 3.96e-09 ***
beta1   9.325e-02   3.556e-02    2.622  0.00873 **
beta2   6.551e-01   3.548e-02   18.461  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 3131.107    normalized:  2.529165
```

The main findings obtained from this summary are alpha1, alpha2, alpha 3, beta1, beta2 and log likelihood.

12: Summary of GARCH (3, 3) on log return of Market Value

```
Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(3, 3), data = x)

Mean and Variance Equation:
 data ~ garch(3, 3)
<environment: 0x000001d3e7601e70>
 [data = x]

Conditional Distribution:
 norm

Coefficient(s):
         mu       omega      alpha1      alpha2      alpha3       beta1       beta2
0.00163931  0.00002770  0.00000001  0.03836586  0.21801565  0.00000001  0.62817220
     beta3
0.09672541

Std. Errors:
 based on Hessian

Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu      1.639e-03   4.802e-04    3.414 0.000641 ***
omega   2.770e-05   3.193e-06    8.675  < 2e-16 ***
alpha1  1.000e-08   1.300e-02    0.000 0.999999
alpha2  3.837e-02   1.053e-02    3.644 0.000268 ***
alpha3  2.180e-01   2.146e-02   10.158  < 2e-16 ***
beta1   1.000e-08         NaN      NaN      NaN
beta2   6.282e-01   4.034e-02   15.570  < 2e-16 ***
beta3   9.673e-02         NaN      NaN      NaN
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 3133.362    normalized:  2.530987
```

The main findings obtained from this summary are alpha1, alpha2, alpha 3, beta1, beta2, beta3 and log likelihood.

13: Summary of GARCH (1, 2) on log return of each company(s) Share Prices

```
Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu      1.125e-04   8.470e-04    0.133   0.8943
omega   4.321e-05   1.680e-05    2.573   0.0101 *
alpha1  1.520e-01   3.183e-02    4.775  1.8e-06 ***
beta1   3.675e-01   2.577e-01    1.426   0.1537
beta2   4.477e-01   2.404e-01    1.862   0.0626 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' '

Log Likelihood:
 2480.292    normalized:  2.003467

Description:
 Thu Jul 18 15:07:11 2024 by user: MYK


Standardised Residuals Tests:
                                 Statistic     p-Value
 Jarque-Bera Test   R    Chi^2    2.6708265 0.263049443
 Shapiro-Wilk Test  R    W        0.9959147 0.002186255
 Ljung-Box Test     R    Q(10)   27.3013744 0.002333092
 Ljung-Box Test     R    Q(15)   28.0073374 0.021523150
 Ljung-Box Test     R    Q(20)   32.3893368 0.039322337
 Ljung-Box Test     R^2  Q(10)   16.1117197 0.096478647
 Ljung-Box Test     R^2  Q(15)   20.9547802 0.138280341
 Ljung-Box Test     R^2  Q(20)   30.9613010 0.055703424
 LM Arch Test       R    TR^2    17.3706720 0.136177793

Information Criterion Statistics:
      AIC       BIC       SIC      HQIC
-3.998856 -3.978172 -3.998888 -3.991076
```

Thus same working of GARCH (1, 2) model is done for all the remaining 29 company(s) share prices.